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Expandable footprint implant

Abstract

Expandable intervertebral fusion implants, system, and methods. The expandable intervertebral implant is capable of being installed inside an intervertebral disc space to maintain disc spacing and restore spinal stability, thereby facilitating an intervertebral fusion. The implant may include a central drive assembly configured to control left and right side assemblies. The expandable intervertebral implant may be configured to transition from a collapsed configuration having a first width and a first height to an expanded configuration having a second width and a second height.

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Primary Examiner: Weiss; Jessica

Background/Summary

(1) This application is a continuation application of U.S. patent application Ser. No. 18/124,095 filed on Mar. 21, 2023, which is incorporated in its entirety herein.

FIELD OF THE INVENTION

(1) The present disclosure generally relates to devices and methods for promoting an intervertebral fusion, and more particularly relates to expandable fusion devices capable of being inserted between adjacent vertebrae to facilitate the fusion process and related systems and methods.

BACKGROUND OF THE INVENTION

(2) A common procedure for handling pain associated with intervertebral discs that have become degenerated due to various factors, such as trauma or aging, is the use of intervertebral fusion devices for fusing one or more adjacent vertebral bodies. Generally, to fuse the adjacent vertebral bodies, the intervertebral disc is first partially or fully removed. An intervertebral fusion device is then inserted between neighboring vertebrae to maintain normal disc spacing and restore spinal stability, thereby facilitating an intervertebral fusion.

(3) There are a number of fusion devices and methodologies for accomplishing the intervertebral fusion. These may include solid bone implants, fusion devices which include a cage or other implant mechanism, which may be packed with bone and/or bone growth inducing substances, and expandable implants. For example, posterior implants may be implanted to provide disc height restoration. The implants may be installed between adjacent vertebral bodies in order to fuse the vertebral bodies together, thereby alleviating the associated pain.

(4) There are drawbacks, however, with existing posterior implants including subsidence and sagittal balance issues. As such, there exists a need for fusion devices capable of being installed inside an intervertebral disc space at a minimum height and width with an expandable footprint to address the subsidence issue and an adjustable lordosis to address the sagittal balance issue.

SUMMARY OF THE INVENTION

(5) To meet this and other needs, devices, systems, and methods for performing intervertebral fusion and spine stabilization are provided. In particular, expandable intervertebral implants, for example, for posterior spinal surgery may be used to treat a variety of patient indications. The expandable implants are configured to increase the overall footprint size after being inserted into the disc space while also adjusting the lordosis and overall height. The in-situ expandable footprint or surface area is configured to address the subsidence issue and in-situ adjustable lordosis is configured to address the sagittal balance issue. The intervertebral implant may be combined with a posterior stabilization system, for example, including pedicle screws and spinal rod(s) to further stabilize the spine.

(6) According to one embodiment, an expandable intervertebral implant includes a central drive assembly and left and right side assemblies. The central drive assembly includes a central drive screw positioned through a front plate, the central drive screw threadedly engaged to a central actuator, the central actuator coupled to a threaded sleeve, and the threaded sleeve positioned through a rear plate and threadedly engaged with a drive nut. The left and right side assemblies each include upper and lower endplates, a side actuator, and a front ramp. The side actuator and front ramp are slidably engaged with the upper and lower endplates. Rotation of the drive nut expands the implant in width and rotation of the drive screw expands the implant in height.

(7) The expandable intervertebral implant may include one or more of the following features. The drive nut may pull the threaded sleeve and central actuator toward the rear plate and push the left and right side portions outwards in width. The drive screw may pull the front ramp toward the central actuator which then expands the upper and lower endplates in height. The central actuator may include a tubular body with an inner bore and a pair of opposite wings configured to mate with the side actuators. The rear plate may include a pair of female horizontal ramps defined into top and bottom surfaces of the rear plate, and the side actuator may include a pair of horizontal male ramps configured to interface with the female horizontal ramps of the rear plate. The front plate may include a pair of female horizontal ramps defined into top and bottom surfaces of the front plate, and the front ramp may include a pair of male horizontal ramps configured to interface with the female horizontal ramps of the front plate.

(8) According to one embodiment, an expandable intervertebral implant includes a front plate having a central through bore, a central drive screw having an enlarged head and a threaded shaft positioned through the bore in the front plate, a central actuator having a tubular body with a through bore and a pair of opposed wings, the threaded shaft of the central drive screw threadedly engaged within one end of the bore of the central actuator, a rear plate having a central through bore, a threaded sleeve positioned through the bore in the rear plate and threadedly engaged with an opposite end of the bore of the central actuator, a drive nut threadedly engaged with the threaded sleeve, and left and right side assemblies each including upper and lower endplates, a side actuator, and a front ramp. The side actuator and front ramp are slidably engaged with the upper and lower endplates, the side actuator is slidably engaged with the rear plate and central actuator, and the front ramp is slidably engaged with the front plate.

(9) The expandable intervertebral implant may include one or more of the following features. The left and right side assemblies may have a laterally collapsed configuration having a first width and a laterally expanded configuration having a second width, and the left and right side assemblies may have a vertically collapsed configuration having a first height and a vertically expanded configuration having a second height. The pair of opposed wings may each define a female ramp configured to receive a corresponding male ramp from the side actuator. The pair of opposed wings may be angled distally toward the front plate. The rear plate may include a cylindrical ring projecting proximally. The cylindrical ring may define an outer threaded surface. The threaded sleeve may include a first proximal exterior threaded section configured to interface with inner threads in a bore of the drive nut, and a second distal exterior threaded section configured to

interface with inner threads in the bore of central actuator. The central drive screw may be retained in the front plate with a retaining ring. The drive nut may be retained in the rear plate with a retaining sleeve.

(10) According to another embodiment, an expandable intervertebral implant includes a central drive assembly and a side assembly. The central drive assembly includes a front plate, a central drive screw, a central actuator, a threaded sleeve, a rear plate, and a drive nut aligned along a central longitudinal axis. The central drive screw is retained in the front plate. The central actuator is threadedly engaged with the central drive screw and the threaded sleeve. The threaded sleeve is retained in the rear plate and threadedly engaged with the drive nut. The side assembly includes upper and lower endplates, a side actuator, and a front ramp. The side actuator and front ramp are slidably engaged with the upper and lower endplates. The side actuator is slidably engaged with the rear plate and central actuator, and the front ramp is slidably engaged with the front plate. When actuated, the drive nut controls width expansion of the implant and the drive screw controls height expansion of the implant.

(11) The expandable intervertebral implant may include one or more of the following features. The central drive screw may extend from a proximal end to a distal end where the distal end includes an enlarged head portion and the proximal end defines an instrument recess. The drive nut may define a central through bore and a proximal face of the drive nut may define an instrument recess different from the instrument recess of the central drive screw. The central drive screw may be located at a distal end of the implant and the drive nut may be located at a proximal end of the implant. When actuated, the drive nut may pull the threaded sleeve and central actuator proximally, thereby causing an expansion in width. When actuated, the central drive screw may pull the front plate toward the central actuator and pull the front ramp toward the side actuator, thereby causing an expansion in height.

(12) According to another embodiment, a method of assembling an expandable intervertebral implant may include one or more of the following steps in any suitable order: (1) placing two front ramps onto a front plate by aligning mating ramps or slides; (2) placing one side actuator onto a rear plate by aligning mating ramps or slides; (3) placing a central actuator onto the side actuator and placing a second side actuator onto the rear plate and the central actuator by aligning the ramps or slides; (4) assembling each of the left and right side assemblies by placing a lower endplate onto the side actuator and placing an upper endplate onto the side actuator; (5) placing the front ramp into both the lower and upper endplates; (6) inserting a threaded sleeve through the rear plate and into the central actuator; (7) placing a friction ring onto the drive nut; (8) threading the drive nut onto the threaded sleeve; (9) pressing a retaining sleeve into the rear plate to secure the drive nut; (10) placing a friction ring onto a central drive screw; (11) inserting the central drive screw through the front plate and threading into the central actuator; and (12) placing a retaining ring into the front plate to secure the central drive screw.

(13) According to another embodiment, a spinal fixation system includes an expandable intervertebral implant, a bone fastener, and a spinal rod attachable to the bone fastener. The expandable intervertebral implant comprising a central drive assembly and left and right side assemblies, the central drive assembly including a front plate, a central drive screw, a central actuator, a threaded sleeve, a rear plate, and a drive nut, and the left and right side assemblies each including upper and lower endplates, a side actuator, and a front ramp, wherein rotation of the drive nut is configured to expand the implant in width and rotation of the drive screw is configured to expand the implant in height.

(14) The spinal fixation system may include one or more of the following features. The expandable intervertebral implant may be configured to be installed through a transforaminal lumbar interbody fusion procedure. The bone fastener may include a threaded shaft and a tulip head. The bone fastener may be a polyaxial bone screw. The bone fastener may be a pedicle screw. The plurality of bone fasteners may be configured to be installed in vertebrae adjacent to the expandable

intervertebral implant. The system may include a set of screws and rods configured to stabilize the spine.

(15) According to another embodiment, a spinal fixation system includes an expandable intervertebral implant comprising a central drive assembly and left and right side assemblies, and a posterior stabilization system. The central drive assembly includes a front plate having a central through bore, a central drive screw having an enlarged head and a threaded shaft positioned through the bore in the front plate, a central actuator having a tubular body with a through bore and a pair of opposed wings, the threaded shaft of the central drive screw threadedly engaged within one end of the bore of the central actuator, a rear plate having a central through bore, a threaded sleeve positioned through the bore in the rear plate and threadedly engaged with an opposite end of the bore of the central actuator, and a drive nut threadedly engaged with the threaded sleeve. The left and right side assemblies each include upper and lower endplates, a side actuator, and a front ramp.

(16) The spinal fixation system may include one or more of the following features. The posterior stabilization system may include a plurality of bone fasteners including a threaded shaft and a tulip head. The plurality of bone fasteners may be configured to be inserted into pedicles of adjacent vertebrae. The posterior stabilization system may include a spinal rod affixed to the tulip heads of the bone fasteners to stabilize a portion of the spine. Actuation of the drive nut may be configured to expand the intervertebral implant in width. Actuation of the central drive screw may be configured to expand the intervertebral implant in height in-situ.

(17) According to yet another embodiment, a method of spinal fixation may include one or more of the following steps in any suitable order: (1) inserting an expandable intervertebral implant into a disc space between adjacent vertebrae, the expandable implant comprising a central drive assembly and left and right side assemblies, the central drive assembly includes a front plate, a central drive screw, a central actuator, a threaded sleeve, a rear plate, and a drive nut, and the left and right side assemblies each include upper and lower endplates, a side actuator, and a front ramp; (2) expanding the implant in width and/or height; (3) affixing bone screws to pedicles of adjacent vertebrae; and (4) connecting a spinal rod between the bone screws. The method may further include expanding the implant in width by rotating the drive nut and expanding the implant in height by rotating the central drive screw. The implant may be expanded in width to increase an overall footprint size, thereby minimizing a potential for implant subsidence. The implant may be expanded in height in-situ to adjust lordosis. Before inserting the expandable intervertebral implant into the disc space, accessing a posterior aspect of a spine, and performing a facetectomy. A unilateral facetectomy may be performed to allow for visualization and removal of a disc. A first set of bone screws may be affixed to pedicles of the vertebrae, and a first spinal rod is connected therebetween, a second set of bone screw may be affixed to opposite pedicle of the vertebrae, and a second spinal rod is connected therebetween.

(18) According to yet another embodiment, a kit may include a plurality of implants of different sizes and configurations. The kit may further include one or more devices suitable for installing and/or removing the implants and systems described herein, such as insertion devices or drivers; one or more removal devices; and other tools and devices, which may be suitable for surgery.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

(1) The present embodiments will become more fully understood from the detailed description and the accompanying drawings, wherein:

(2) FIGS. 1A-1C illustrate perspective views of an expandable implant in a collapsed position, expanded in width, and expanded in width and height, respectively, according to one embodiment;

- (3) FIG. 2 shows an exploded view of the expandable implant of FIGS. 1A-1C;
- (4) FIG. 3 shows an exploded view of a portion of the central drive assembly including a central drive screw insertable into a front plate and configured to be retained by a retaining ring according to one embodiment;
- (5) FIG. 4 shows an exploded view of another portion of the central drive assembly including a central drive nut, rear plate, threaded sleeve, and retaining sleeve for securing the drive nut in the rear plate according to one embodiment;
- (6) FIG. 5 shows the central drive assembly including the rear plate attached to the drive sleeve, the drive sleeve attached to the central actuator, the central actuator attached to the central drive screw, and the central drive screw attached to the front plate according to one embodiment;
- (7) FIG. 6 shows a side assembly including the top endplate assembled onto the side actuator and the front ramp and a lower endplate configured to be placed onto the side actuator and front ramp during assembly;
- (8) FIG. 7 shows a cross-sectional view of the ramps of the upper and lower endplates engaged with the side actuator and front ramp according to one embodiment;
- (9) FIG. 8 shows the side actuator of the side assemblies according to one embodiment;
- (10) FIG. 9 shows the front ramp of the side assemblies according to one embodiment;
- (11) FIGS. 10A-10C illustrate the side actuators slidably engaged with the rear plate in a collapsed position and expanded in width, respectively, and a close-up view of the interaction between the side actuator and the rear plate according to one embodiment;
- (12) FIGS. 11A-11B illustrate the front ramps slidably engaged with the front plate in a collapsed position and expanded in width, respectively, according to one embodiment;
- (13) FIGS. 12A-12B show rear and perspective views, respectively, of the implant expanded in width according to one embodiment;
- (14) FIGS. 13A-13B show the implant fully expanded in width and fully expanded in width and height, respectively, according to one embodiment (one side assembly including one set of endplates, side actuator, and front ramp are omitted for clarity); and
- (15) FIG. 14 illustrates a spinal construct including the expandable implant in combination with pedicle screws and attached spinal rod for stabilizing the posterior spine according to one embodiment.

DETAILED DESCRIPTION

(16) Embodiments of the disclosure are generally directed to devices, systems, and methods for intervertebral fusion and spine stabilization. Specifically, expandable implants are configured to increase the overall footprint size after being inserted into the disc space while also adjusting the lordosis and overall height. The expandable implants may include one or more side assemblies configured to expand in width and in height. In doing so, the expansion addresses sagittal balance correction and subsidence issues. The intervertebral implant may be combined with a posterior stabilization system, for example, including pedicle screws and spinal rod(s) to further stabilize the spine.

(17) A spinal fusion is typically employed to eliminate pain caused by motion of degenerated disc material. Upon successful fusion, a fusion device becomes permanently fixed within the intervertebral disc space. The expandable fusion device may be positioned between adjacent vertebral bodies in a collapsed position. The expandable fusion device is configured to expand in width and subsequently in height. The fusion device engages the endplates of the adjacent vertebral bodies and, in the installed position, maintains desired intervertebral disc spacing and restores spinal stability, thereby facilitating the intervertebral fusion.

(18) Minimally invasive surgery (MIS) may be used to preserve muscular anatomy by only causing disruption where necessary. The benefit of the MIS surgical approach is that it can reduce post-operative pain and improve recovery time for patients. In one embodiment, the expandable fusion device can be configured to be placed down an endoscopic tube and into the surgical target site. By

way of example, the surgical site may be an intervertebral disc space situated between two adjacent vertebrae. Although particularly suited for use in a transforaminal lumbar interbody fusion (TLIF), it will be readily appreciated by those skilled in the art that the implant may be employed in any number of suitable orthopedic approaches and procedures, including but not limited to, anterior, posterior, lateral, anterolateral, or posterolateral approaches to the lumbar spine, cervical spine, or thoracic spine, as well as any non-spine application, such as treatment of bone fractures and the like.

(19) Components of all of the devices disclosed herein may be manufactured of any suitable materials including metals (e.g., titanium), metal alloys (e.g., stainless steel, cobalt-chromium, and titanium alloys), ceramics, plastics, plastic composites, or polymeric materials (e.g., polyether ether ketone (PEEK), polyphenylene sulfone (PPSU), polysulfone (PSU), polycarbonate (PC), polyetherimide (PEI), polypropylene (PP), polyacetals, or mixtures or co-polymers thereof), and/or combinations thereof. In some embodiments, the devices may include radiolucent and/or radiopaque materials. The components can also be machined and/or manufactured using any suitable techniques (e.g., 3D printing).

(20) Turning now to the drawing, where like reference numerals refer to like elements, FIGS. 1A-1C illustrate an expandable fusion device or implant **10** according to one embodiment. The expandable fusion device **10** may include left and right side portion assemblies **12**, **14** configured to expand in width to increase the overall footprint of the device **10** and expand in height to correct disc height restoration, lordosis, and/or sagittal balance. The implant **10** may be suitable for a transforaminal lumbar interbody fusion (TLIF) through a posterior approach or other suitable surgical procedure.

(21) The expandable fusion device **10** extends along a central longitudinal axis A between front and rear ends of the device **10**. FIG. 1A shows the expandable fusion device **10** in a fully collapsed configuration with the left and right side portions **12**, **14** collapsed in both width and height. FIG. 1B shows the expandable fusion device **10** in an expanded configuration with the left and right side portions **12**, **14** expanded in width. FIG. 1C shows the expandable fusion device in a fully expanded configuration with the left and right side portions **12**, **14** expanded in width and in height. It should be understood that references to the front and rear ends and left and right side portions **12**, **14** are described with respect to the direction of placement into an intervertebral disc space with the front of the expandable fusion device **10** placed into the disc space first, followed by the rear of the expandable fusion device **10**. These and other directional terms may be used herein for descriptive purposes and do not limit the orientation(s) in which the devices may be used.

(22) With emphasis on the exploded view in FIG. 2, the implant **10** includes a first half or left side portion **12**, a second half or right side portion **14**, and a central drive assembly **16**. The left and right side portion assemblies **12**, **14** may each include upper and lower endplates **20**, **22**, a front ramp **24**, and a side actuator **26**. The central drive assembly **16** includes a central drive screw **30**, a front plate **32**, a central actuator **34**, a rear plate **36**, a central threaded sleeve **38**, and a central drive nut **40** aligned along the central longitudinal axis A. The left and right side portions **12**, **14** are expanded in width by the central drive nut **40**. The upper and lower endplates **20**, **22** are expanded in height by the central drive screw **30**.

(23) In particular, the left and right side portions **12**, **14** are controllable by the central drive nut **40** which is attached to the central threaded sleeve **38** and central actuator **34**. The drive nut **40** pulls the threaded sleeve **38** and central actuator **34** toward the rear plate **36** and pushes the left and right side portions **12**, **14** outwards with the uses of ramps or slides **70**, **102**, **132**, **154**, **156**, **170**. Once the left and right side portions **12**, **14** are fully expanded in width, the front plate **32** is controllable by the central drive screw **30**. The central drive screw **30** pulls the front plate **20** toward the central actuator **34** while also pulling the front ramps **24** toward the side actuators **26**. Each side assembly **12**, **14** has upper and lower endplates **20**, **22**, front ramp **24**, and side actuator **26**. The front ramp **24** is actuated while the drive screw **30** is turned. This pulls the front ramp **24** toward the side actuator

26 which then expands the top and bottom endplates **20**, **22** up and down with mating ramps **142**, **160**, **168** on the front ramps **24** and side actuators **26**.

(24) As best seen in FIG. **3**, the central drive screw **30** extends from a proximal end **44** to a distal end **46**. The distal end **46** may include an enlarged head portion **48** configured to be received in a bore **62** defined through the front plate **32**. The enlarged head **48** may include a wide cylindrical head with a smooth outer surface at the distal end of the central drive screw **30**. The proximal end **44** of the drive screw **30** may define an instrument recess **50** configured to receive an instrument, such as a driver, to rotate or actuate the drive screw **30**. The instrument recess **50** may include a tri-lobe, hex, star, or other suitable recess configured to engage with a driver instrument to apply torque to the drive screw **30**. The drive screw **30** may include a shaft with an exterior threaded portion **52** extending along its length. The exterior threads **52** may extend from the proximal end **44** to a location near the bottom of the enlarged head **48**. The drive screw **30** is receivable through the bore **62** in the front plate **32** such that the enlarged head portion **48** of the drive screw **30** is receivable in the front plate **32**. An optional friction ring **56**, such as a polyether ether ketone (PEEK) ring, may be assembled onto the drive screw **30**, for example, below the enlarged head **48**, to increase friction or drag on the drive screw **30** during rotation.

(25) The central drive screw **30** is inserted into the front plate **32** and may be retained within the front plate **32** using a retaining ring **58**. The drive screw **30** controls height expansion of the implant **10** when actuated. The retaining ring **58** may include a split ring with a plurality of inner and/or outer teeth **60** or various reliefs to allow the retaining ring **58** to compress and enter the bore **62** of the front plate **32** and engage an internal groove. The retaining ring **60** may include two slots, for example, to be engaged with an instrument to aid insertion and removal of the retaining ring **58**. When the retaining ring **58** is positioned around the drive screw **30** and within the bore **62** in the front plate **32**, the teeth **58** are configured to engage with the central drive screw **30**, thereby locking the screw **30** in position in the plate **32**.

(26) The front plate **32** defines a central bore **62** therethrough configured to receive drive screw **30**. The axis of bore **62** is aligned along central longitudinal axis A. The front plate **32** includes an upper face or top surface **64** and an opposite lower face or bottom surface **66** connected by side surfaces **68**. The top and bottom surfaces **64**, **66** may include one or more slides or ramps **70** configured to interface with corresponding ramps **170** on the front ramps **24** of the left and right side portions **12**, **14**. Slides or ramps **70** may include horizontal ramps defining female channels or grooves configured to receive the mating male counterparts **170** of the front ramps **24**. It will be appreciated, however, that the female/male configurations may be reversed or may include other suitable ramp interactions, sliding features, or mating components to provide lateral expansion of the left and right side portions **12**, **14**.

(27) In one embodiment, the front plate **32** may include a first pair of horizontal ramps **70** defined into the top **64** of the front plate **32** and a second pair of horizontal ramps **70** defined into the bottom **66** of the front plate **32**. Each of the ramps **70** may be aligned along distinct horizontal planes. In this manner, each ramp **70** has a constant depth along its length such that one of the female horizontal ramps has a depth greater than the other female horizontal ramp. For example, a first ramp **70** defined along the top **64** of the front plate **32** may be positioned along one given horizontal plane lower or higher relative to the other ramp **70** defined along the top **64** of the front plate **32**. The horizontal ramps **70** may be angled, diagonal, or slanted such that one end of each ramp **70** begins at a side **68** of the front plate **32** and extend centrally in a direction toward the central actuator **34** with the ramps **70** leading toward one another. Although described as horizontal ramps to provide for horizontal or lateral expansion of the left and right side assemblies **12**, **14**, it will be appreciated that the horizontal ramps **70** may be sloped, slanted, or otherwise configured off axis to provide for a different trajectory or type of expansion.

(28) As best seen in FIG. **4**, drive nut **40** is inserted into the rear plate **36** and may be retained using retaining sleeve **74**. The drive nut **40** controls width expansion of the implant **10** when actuated.

The drive nut **40** may have a generally cylindrical body that extends from a proximal end **76** to a distal end **78**. The proximal face **76** may define one or more indentions or notches to form an instrument recess **80** configured to interface with an instrument, such as a driver, to rotate or actuate the drive nut **40**. The instrument recess **80** may define a slotted, splined, tri-lobe, hex, star, other suitable recess, or a portion thereof configured to engage with a driver instrument to apply torque to the drive nut **40**. The instrument recess **80** of the drive nut **40** may be of a different type than the instrument recess **50** of the drive screw **30** so the user can easily discern which component is intended to be actuated. In this manner, the instrument recess **80** is configured to engage with a driver instrument to apply torque to the drive nut **40**, thereby driving width expansion of the left and right side assemblies **12, 14**.

(29) The drive nut **40** defines a central through bore **82** such that another instrument is able to access the distal drive screw **30**. The axis of through bore **82** may aligned along central longitudinal axis A. The drive nut **40** is receivable within the bore **88** in the rear plate **36** such that the body of the drive nut **40** is receivable in the rear plate **36**. An optional friction ring **84**, such as a polyether ether ketone (PEEK) ring, may be assembled onto the drive nut **40**, for example, near distal end **78**, to increase friction or drag on the drive nut **40** during rotation. The retaining sleeve **74** is pressed into the rear plate **36**, thereby securing the drive nut **40** in the rear plate **36**. The retaining sleeve **74** may include a ring or band sized and dimensioned to fit around the drive nut **40** and within the projection **96** of the rear plate **36**.

(30) The rear plate **36** defines a central bore **88** therethrough configured to receive drive nut **40** and threaded sleeve **38**. The axis of bore **88** is aligned along central longitudinal axis A. The rear plate **36** includes an upper face or top surface **90** and an opposite lower face or bottom surface **92** connected by side surfaces **94**. A projection **96** may extend proximally defining a substantially cylindrical ring with an outer threaded portion **98**, for example, configured to threadedly engage with an implant insertion instrument. A threaded connection to the rear plate **36** provides for a rigid connection to the insertion instrument. The top and bottom surfaces **90, 92** may include one or more slides or ramps **102** configured to interface with corresponding ramps **154** on the side actuators **26** of the left and right side portions **12, 14**.

(31) The rear plate **36** includes one or more slides or ramps **102** configured to interface with corresponding ramps **154** on the actuators **26** of the left and right side portions **12, 14**. For example, near the distal end of the rear plate **36**, the rear plate **36** may include a pair of ramps **102** defined into each of the top and bottom surfaces **90, 92** of the rear plate **36**. Similar to ramps **70**, ramps **102** may be horizontal ramps aligned along one or more horizontal planes. For example, one of the pair of ramps **102** may be positioned along one given horizontal plane lower or higher relative to the other ramp **102** along another given horizontal plane. In other words, each ramp **102** has a constant depth along its length such that one ramp **102** has a depth greater than the other ramp **102**. The horizontal ramps **102** may be angled, diagonal, or slanted such that one end of the ramp **102** starts at a side **94** of the rear plate **36** and extends toward the center of the rear plate **36** with the ramps **102** leading toward one another. The horizontal ramps **102** may define female channels or grooves configured to receive the mating male counterparts **154** of the actuators **26**. It will be appreciated, however, that the female/male configurations may be reversed or may include other suitable ramp interactions, sliding features, or mating components to provide lateral expansion of the left and right side portions **12, 14**.

(32) The rear plate **36** may include one or more instrument recesses or slots **104, 106** configured to be engaged by an instrument, such as an insertion instrument, or allow for access to the implant **10**. For example, the side faces **94** of the rear plate **36** may each include side recesses or instrument slots **104** configured to receive a graft delivery device. For example, opposite sides **94** of the rear plate **36** may include two opposed semi-circular recesses **104** configured to allow the graft delivery device to enter the central portion of the implant **10** once fully expanded in width and/or height. Bone graft or similar bone growth inducing material can be introduced within and/or around the

fusion device **10** to further promote and facilitate the intervertebral fusion. In addition, one or more recesses **106** may be positioned about the proximal face of the rear plate **36**. These circular recesses **106** may be aligned at the four corners of the plate **36**. The quadruple recesses **106** may help to align an instrument, such as insertion instrument.

(33) The drive nut **40** is threaded onto threaded sleeve **38**. The threaded sleeve **38** is positioned through bore **88** of rear plate **36** and threadedly engaged to drive nut **40**. The threaded sleeve **38** is a cylindrical sleeve or shaft sleeve defining a central through bore **114** between proximal end **110** and distal end **112** of the sleeve **38**. The axis of the central bore **114** aligns with the central longitudinal axis A. An outer surface of the sleeve **38** includes exterior threaded sections **116**, **118**. A first proximal exterior threaded section **116** is configured to interface with inner threads in bore **82** of drive nut **40**. A second distal exterior threaded section **118** is configured to interface with inner threads in bore **124** of central actuator **34**. The threaded sections **116** may include helically wound thread forms having any suitable lead, pitch, handedness, angle, diameters, etc. The separate thread sections **116**, **118** may have the same or different thread types. The thread sections **116**, **118** may be separated by a non-threaded or smooth section **120**. The proximal threaded section **116** may be longer than the distal threaded section **118** such that the non-threaded section **120** is located closer to the distal end **112** of the sleeve **38**. For example, the length of the proximal threaded section **116** may be double, triple, quadruple, or greater than the length of the distal threaded section **118**.

(34) Turning now to FIG. 5, the fully assembled central drive assembly **16** is shown. The threaded sleeve **38** is attached to central actuator **34**. The central actuator **34** includes a tubular body with an inner bore **124** extending from a proximal end **126** to a distal end **128**. The inner bore **124** of the central actuator **34** is internally threaded to allow for threaded engagement with the distal threaded section **118** of the threaded sleeve **38** and the threaded shaft **52** of the drive screw **30**. The central actuator **34** includes a pair of opposite wings **130** extending from an outer surface of the tubular body. Each wing **130** may terminate at a distal free end. The wings **130** may be angled such that the distal free ends of the wings **130** point toward the front plate **32**. Each wing **130** may define one or more slides or ramps **132** configured to mate with corresponding surfaces on the side actuators **26**. The ramps **132** may be angled, diagonal, chamfered, or slanted such that one end of the ramp **132** starts at the body of the central actuator **34** and extends to the free end of the respective wing **130**. The ramps **132** may define female channels or grooves configured to receive the mating male counterparts **156** of the side actuators **26**. The female ramps **132** may extend openly along a proximal face of each wing **130**. It will be appreciated that the female/male configurations and locations may be reversed or changed to include other suitable ramp interactions, sliding features, or mating components to help provide lateral expansion of the left and right side portions **12**, **14**.

(35) When the central drive assembly **16** is assembled, the proximal end **110** of the threaded sleeve **38** is positioned through the bore **88** in the rear plate **36**. The drive nut **40** is threaded onto the proximal threaded section **116** of the threaded sleeve **38**. The distal threaded section **118** of the threaded sleeve **38** is threaded into the central actuator **34**. The actuation of the drive nut **40** is configured to push and pull the threaded sleeve **38** and central actuator **34** relative to the rear plate **36** and expand the implant **10** in width once fully assembled. The drive screw **30** is positioned through the bore **62** in the front plate **32**. The threaded shaft **52** of the drive screw **30** is threaded into the distal end **128** of the central actuator **34**, which is attached to the threaded sleeve **38**. The actuation of the drive screw **30** is configured to push and pull the front plate **32** and front ramps **24** relative to the central actuator **34** and side actuators **26** and expand the implant **10** in height once fully assembled.

(36) Turning now to FIGS. 6 and 7, the left and right side assemblies **12**, **14** are each assembled by placing the upper and lower endplates **20**, **22** onto side actuator **26** and front ramp **24**. The left and right side portion assemblies **12**, **14** may each include upper and lower endplates **20**, **22** configured to expand away from one another to increase the vertical height of the expandable fusion device **10**. The upper and lower endplates **20**, **22** may be the same or mirror images of one another. Although

described with reference to assembly **12** and upper endplate **20**, the discussion herein applies equally to assembly **14** and lower endplate **22**. The upper endplate **20** includes an upper or outer facing surface **136** configured to interface with the vertebral endplate(s) of the adjacent vertebral bodies when implanted in the disc space. The outer surface **136** may include a plurality of teeth, ridges, roughened surfaces, keels, gripping or purchasing projections, or other friction increasing elements configured to retain the device **10** in the disc space. For example, the endplates **20**, **22** may be 3D printed using additive manufacturing to provide a natural roughened surface to promote bone on growth or may be machined and blasted to achieve a roughened surface.

(37) The upper endplate **20** includes a lower or inner facing surface **138** and one or more side walls **140** defining one or more slides or ramps **142** configured to interface with corresponding ramps **160**, **168** on the actuator **26** and front ramp **24**, respectively. For example, the upper endplate **20** may define a pair of sidewalls **140** forming a gap or channel **144** therebetween. The channel **144** may be generally U-shaped, J-shaped, C-shaped, or another suitable configuration. At least three pairs of ramps **142** may be defined within the channel **144** of the endplate **20**. The ramps **142** may be vertical ramps aligned along one or more vertical planes. In one embodiment, all three pairs of vertical ramps **142** may be aligned along the same plane. Although vertically-oriented, the vertical ramps **142** may be angled, diagonal, or sloped to increase the vertical height of the endplates **20**, **22**.

(38) In one embodiment, two pairs of vertical ramps **142** interfaced with the side actuator **26** may be angled in one direction and a third pair of vertical ramps **142** interfaced with the front ramp **24** may be angled in an opposite direction. For example, the distal-most vertical ramps **142**, near front ramp **24**, may be sloped such that they point toward the front ramp **24** as they extend along the side walls **140** from the inner surface **138** toward the outer surface **136**. Similarly, the proximal-most vertical ramps **142**, near the side actuator **26**, and centrally located vertical ramps **142**, may be sloped such that they point toward the side actuator **26** as they extend along the side walls **140** from the inner surface **138** toward the outer surface **136**. The proximal-most vertical ramps **142** and central vertical ramps **142** may be aligned in parallel with the same degree of slope or may have different degrees of slope.

(39) The vertical ramps **142** may define female channels or grooves configured to receive the mating male counterparts **160**, **168** of the side actuator **26** and front ramp **24**, respectively. It will be appreciated that the female/male configurations may be reversed or may include other suitable ramp interactions, sliding features, or mating components to provide vertical expansion of the left and right side portions **12**, **14**. Although described as vertical ramps to provide for vertical expansion of the left and right side assemblies **12**, **14**, it will be appreciated that the vertical ramps **142** may be sloped, slanted, or otherwise configured off axis to provide for a different trajectory or type of expansion.

(40) One or more openings **146** may extend vertically through the body of the endplate **20**, **22**. In the collapsed position, as shown in FIG. **1A**, ramp portions **160** of the side actuators **26** may be received through the openings **146**. Similarly, when expanded in width, as shown in FIG. **1B**, the ramps **160** of the side actuators **26** are receivable through the openings **146** in the endplates **20**, **22**. In the vertically expanded position, as shown in FIG. **1C**, the openings **146** may be open and free to receive bone-graft or other suitable bone forming material. One or more openings or slots **148** may extend horizontally through the sidewalls **140** of the endplate **20**, **22**. The slots **148** may be provided between the distal-most and central vertical ramps **142** and near the inner surface **138** of the endplate **20**, **22**. In the collapsed position, as shown in FIG. **1A**, slots **148** from the upper and lower endplates **20**, **22** may align such that the wings **130** of the central actuator **34** is positionable through the openings **148**. When expanded in width, as shown in FIG. **1B**, the side assemblies **12**, **14** expand outward and away from one another and the openings **148** or a portion thereof are vacant.

(41) Turning now to FIG. **8**, the left and right side portion assemblies **12**, **14** may include first and

second side actuators **26** positioned between the upper and lower endplates **20**, **22** of the left and right side portions **12**, **14**, respectively. The actuator **26** may include a body extending along a central axis **A1** from a proximal end **150** to a distal end **152**. Central axis **A1** may be generally parallel with central longitudinal axis **A**. The proximal end **150** may define one or more horizontal slides or ramps **154** configured to engage with the horizontal ramps **102** of the rear plate **36**. The side actuator **26** may define a pair of top and bottom horizontal ramps **154** pointing inwardly toward one another. The horizontal ramps **154** may have an angled, diagonal, or slanted surface in a manner complimentary to the ramps **102** of the rear plate **36**. In particular, the horizontal ramps **154** may define male projections configured to enter the female counterparts **102** of the rear plate **36**. The ramp **154** may define a limiter **176** to end movement of the mating slides or ramps **102**, **154**. For example, the limiter **176** may include a widened ramp surface, a projection or protrusion, a bevel or dovetail, etc. configured to reduce or limit the relative movement between the ramps **102**, **154**.

(42) The distal end **152** of the side actuator **26** may include ramp **156** configured to engage with the corresponding ramp **132** on the wing **130** of the central actuator **34**. The side actuator **26** may define upper and lower grooves **158** to form a single male ramp **156** pointing distally. The male ramp **156** may have an angled, diagonal, or slanted surface in a manner complimentary to the female ramp **132** on wing **130** of the central actuator **34**. This slidable interface **132**, **156** may be configured to help guide the outward expansion in width of the first and second side assemblies **12**, **14**.

(43) The actuator **26** includes a plurality of ramps **160** configured to engage with the endplates **20**, **22**. The actuator **26** may define a plurality of vertical ramps **160** configured to engage with the vertical ramps **142** of the endplates **20**, **22**. The actuator **26** may define a first set of vertical ramps **160** slanted upwardly toward the proximal end **150** or downwardly toward the distal end **152** and a second set of vertical ramps **160** slanted upwardly toward the distal end **152** or downwardly toward the proximal end **150** of the actuator **26**. Each set of ramps **160** may define a pair of male projections configured to enter the female counterparts **142** of the endplates **20**, **22**. The vertical ramps **160** may have an angled, diagonal, or sloped surface in a manner complimentary to the corresponding ramps **142** of the endplates **20**, **22**.

(44) As shown in FIG. **9**, the left and right side portions **12**, **14** may each include front ramp **24** configured to expand the distal or front ends of the upper and lower endplates **20**, **22**. The front ramp **24** may include a body extending from a proximal end **164** to a distal end **166**. The proximal end **164** may define one or more vertical ramps **168** configured to engage with the vertical ramps **142** of the endplates **20**, **22**. The front ramp **24** may define first vertical ramps **168** sloped downwardly toward the proximal end **164** and a second vertical ramp **168** sloped upwardly toward the proximal end **164** of the front ramp **24**. Each of ramps **168** may define a pair of opposed male projections configured to enter the female counterparts **142** in the channel **144** of the endplates **20**, **22**. The vertical male ramps **168** may have an angled, diagonal, or sloped surface in a manner complimentary to the female ramps **142** of the endplates **20**, **22**.

(45) The distal end **166** of the front ramp **24** may define one or more horizontal ramps **170** configured to engage with the horizontal ramps **70** of the front plate **32**. The front ramp **32** may define a pair of top and bottom horizontal ramps **170** separated by a gap and pointing inwardly toward one another. The horizontal ramps **170** may have an angled, diagonal, or slanted surface in a manner complimentary to the ramps **70** of the front plate **32**. In particular, the horizontal ramps **170** may define male projections configured to enter the female counterparts **70** of the front plate **32**. Similar to ramps **154**, the ramps **170** may define a limiter **176** to end movement of the mating slides or ramps **70**, **170**. For example, the limiter **176** may include a widened ramp surface, a projection or protrusion, a bevel or dovetail, etc. configured to reduce or limit the relative movement between the ramps **70**, **170**.

(46) FIGS. **10A-10C** show the side actuators **26** slidably engaged with the rear plate **36** in collapsed

and expanded positions, respectively. The side actuators **26** slide onto the rear plate **36** by aligning keying features that control expansion. In FIG. **10A**, the actuators **26** are engaged with the rear plate **36** and collapsed onto one another. A recessed surface **162** on top of the actuator **26** sized and dimensioned to receive the other actuator **26** permits the actuators **26** to nest together, thereby providing a small footprint for insertion. It will be appreciated that a corresponding recessed surface **162** may be provided on the bottom of the opposite actuator **26** to provide for a complimentary fit. As best seen in FIG. **10C**, one horizontal ramp **102** is positioned deeper than another horizontal ramp **102** to further facilitate this nesting configuration of the adjacent side actuators **26**. The horizontal male ramps **154** of the side actuators **26** slidably interface with the horizontal female ramps **102** of the rear plate **36**, thereby permitting lateral or horizontal movement of the left and right side assemblies **12**, **14**. The side actuators **26** may have features that engage with the mating rear plate **36**, such as limiters **176**, that limit the amount of translation while expanding in width. When expanded laterally in width, the side actuators **26** slide outward and away from one another, thereby increasing the width of the implant **10**. FIG. **10B** shows the side actuators **26** fully expanded in width relative to the rear plate **32**.

(47) Front ramps **24** and front plate **32** utilize a similar sliding interface as actuators side **26** and rear plate **36**. FIGS. **11A-11B** show the front ramps **24** slidably engaged with the front plate **32** in collapsed and expanded positions, respectively. Front ramps **24** slide onto the front plate **32** by aligning keying features that control expansion. In FIG. **11A**, the front ramps **24** are engaged with the front plate **32** and collapsed onto one another. A recessed surface **172** on top of one front ramp **24** sized and dimensioned to receive the other front ramp **24** permits the front ramps **24** to nest together, thereby providing a small footprint for insertion. It will be appreciated that a corresponding recessed surface **172** may be provided on the bottom of the opposite front ramp **24** to provide for a complimentary fit. Similar to rear plate **36**, one horizontal ramp **70** may be positioned at a depth greater than the other horizontal ramp **70** to further facilitate this nesting configuration of the front ramps **24**. The horizontal ramps **170** of the front ramps **24** slidably interface with the horizontal ramps **70** of the front plate **32**. The front ramps **24** may also have features that engage with the mating front plate **32**, such as limiters **176**, that limit the amount of translation while expanding in width. In FIG. **11B**, the front ramps **24** are fully expanded in width. When expanded in width, the front ramps **24** slide outward and away from one another, thereby increasing the overall width of the implant **10**.

(48) FIGS. **12A-12B** show rear and perspective views, respectively, of the assembled implant **10** when fully expanded in width. The user is able to access the central drive assembly **16** through bore **88** in the rear plate **36**. In particular, the drive nut **40** and drive screw **30** may be independently rotated or turned to actuate the implant **10**. When the drive nut **40** is actuated, the threaded sleeve **38** is drawn proximally and the left and right side portions **12**, **14** expand in width. Once expanded in width, the rear plate **36**, which defines recesses **104**, aligns with corresponding recesses **174** on the side actuators **28** to allow for access to a central portion of the implant **10**. In this manner, a graft delivery device may be used to enter a central open area inside the implant **10** and deliver graft material therein. When the drive screw **30** is actuated, the left and right side portions **12**, **14** expand in height.

(49) FIGS. **13A-13B** show the implant **10** with the left side assembly **12** (one set of endplates **20**, **22**, side actuator **26**, and front ramp **24** and hidden for clarity). The implant **10** may be fully collapsed to be inserted into a disc space, for example, through a posterior approach. The implant **10** may be attached to a multi-component instrument, for example. A first width driving instrument may be used to engage the recess **80** in the drive nut **40** with compatible drive features. When the drive nut **40** is rotated, it pulls or translates the threaded sleeve **38**, central actuator **34**, and front plate **32** proximally. When doing so, the drive nut **40** forces the side actuators **26** and front ramps **24** to translate outward and expand in width. Once the central actuator **34** and rear plate **36** come in contact with each other the drive nut **40** cannot be turned any further. As shown in FIG. **13A**, full

width expansion has been achieved. A second height driving instrument may be used to engage the recess **52** in the drive screw **30** with compatible drive features. When the drive screw **30** is rotated, the drive screw **30** pulls or translates the front plate **32** and front ramps **24** proximally. When doing so, the front ramps **24** translate proximally toward the side actuators **26** and force the endplates **20**, **22** to translate upward and downward from the axial plane. Once the front plate **32** and the central actuator **34** come in contact with each other the drive screw **30** cannot be turned any further. As shown in FIG. **13B**, the implant **10** is fully expanded in width and in height. The amount of height expansion on the left and right side assemblies **12**, **14** may be the same or different. In this manner, the implant **10** is expanded in width for an increased footprint to aid in overall stability and the implant **10** is adjusted in lordosis and height for a precise patient fit.

(50) The implant **10** may have a dual-sided or single-sided width expansion. In one embodiment, the expandable fusion device or implant **10** may be configured such that only one side assembly **12**, **14** expands in width and the other remains stationary. Both of the left and right side portions **12**, **14** may expand in height. In this embodiment, the front and rear plates **32**, **36** may include only a single horizontal ramp on each of the top and bottom faces to engage with a single side actuator **26** and front ramp **24**, respectively. The single side that does not expand laterally outward may incorporate the front ramp features into the front distal plate **32** and the actuator features into the rear proximal plate **36**.

(51) The implant **10** may be assembled as follows. The two front ramps **24** are placed onto the front plate **32** by aligning the slides or ramps **70**, **170**. One side actuator **26** is placed onto rear plate **36** by aligning the slides or ramps **102**, **154**. The central actuator **34** is placed onto the side actuator **26**. The other side actuator **26** is placed onto the rear plate **36** and central actuator **34** by aligning slides or ramps **132**, **156**. The left and right sides **12**, **14** may each be assembled by placing the lower endplate **22** onto the side actuator **26** and placing the upper endplate **20** onto side actuator **26**. The front ramp **24** is placed into both the lower and upper endplates **20**, **22**. The threaded sleeve **38** is inserted into through the rear plate **36** and into central actuator **34** and secured. A friction ring **84** may be placed onto the drive nut **40**. The drive nut **40** is then threaded on threaded sleeve **38** and bottomed out. The retaining sleeve **74** is pressed into rear plate **36**, thereby securing drive nut **40**. A friction ring **56** may be placed onto drive screw **30**. Drive screw **30** is inserted through front plate **32** and threaded into central actuator **34**. Retaining ring **58** may be placed into front plate **32**, thereby securing drive screw **30**.

(52) Turning now to FIG. **14**, a system **210** for intervertebral fusion is shown according to one embodiment. The system **210** may include expandable implant **10**, a plurality of bone fasteners **212**, and a spinal rod **214** connecting the bone fasteners **212**. The expandable fusion device **10** may be surgically implanted into an intervertebral disc space **202** situated between upper and lower vertebrae **204**, **206**. A posterior aspect of the spine may be accessed, for example, through a minimally invasive surgical procedure. In one embodiment, the implant **10** may be implanted using a transforaminal lumbar interbody fusion (TLIF) procedure where a facet joint may be removed to access the disc space **202**. The intervertebral disc or a portion thereof may be removed. The TLIF procedure may include a unilateral facetectomy to allow for visualization and removal of the disc. The implant **10** may be positioned in between the vertebrae **204**, **206** to fuse the spine segment.

(53) The disc level or levels may be further stabilized by inserting bone fasteners **212**, such as pedicle screws, into the vertebrae **204**, **206** above and below the implant **10**, and connecting a spinal rod **214** to the fasteners **212**. The bone fasteners **212** may include bone screws, anchors, clamps, or the like configured to engage bone. In the embodiment shown, the bone fastener **212** is a bone screw, which extends from a proximal end to a distal tip **216**. The proximal end may include an enlarged head receivable within a modular tulip element **218**, such as a polyaxial bone screw. The tulip head **218** is configured to secure spinal rod **214**. Examples of bone fasteners and other implants and rod constructs are described in more detail, for example, in U.S. Pat. No. 10,368,917, which is incorporated by reference herein in its entirety for all purposes. The pedicle screws **212**

may be used with a spinal fusion to add extra support and strength to the fusion while it heals. The pedicle screws **212** may be placed above and below the vertebrae **204, 206** to be fused. Although only a single rod **214** is shown, a bilateral construct may be used with a pair of rods **214** secured laterally along each side of the spine.

(54) Although the invention has been described in detail and with reference to specific embodiments, it will be apparent to one skilled in the art that various changes and modifications can be made without departing from the spirit and scope of the invention. Thus, it is intended that the invention covers the modifications and variations of this invention provided they come within the scope of the appended claims and their equivalents. It is expressly intended, for example, that all components of the various devices disclosed above may be combined or modified in any suitable configuration.

Claims

1. A spinal fixation system comprising: an expandable intervertebral implant comprising a central drive assembly aligned along a central longitudinal axis of the implant and left and right side assemblies defining opposing lateral sides of the implant, the central drive assembly including a front plate at a distal end of the implant, an externally threaded central drive screw, a central actuator having a tubular body with a through bore comprising inner threads and a pair of opposed wings, an externally threaded sleeve, a rear plate, and a drive nut at a proximal end of the implant, wherein the central drive screw is retained in the front plate and is threadedly engaged with the inner threads of the tubular body of the central actuator, the central actuator is threadedly engaged with the externally threaded sleeve, and the externally threaded sleeve is retained in the rear plate and threadedly engaged with the drive nut, and the left and right side assemblies each including upper and lower endplates, a side actuator, and a front ramp, wherein the pair of opposed wings each define a female ramp configured to receive a corresponding male ramp from each side actuator, wherein the rear plate includes a pair of female horizontal ramps defined into top and bottom surfaces of the rear plate, and each side actuator includes a pair of horizontal male ramps configured to interface with the respective female horizontal ramps of the rear plate; wherein the front plate includes a pair of female horizontal ramps defined into top and bottom surfaces of the front plate, and each front ramp includes a pair of male horizontal ramps configured to interface with the respective female horizontal ramps of the front plate, wherein the left and right side assemblies have a laterally collapsed configuration having a first width and a laterally expanded configuration having a second width larger than the first width, and wherein the left and right side assemblies have a vertically collapsed configuration having a first height and a vertically expanded configuration having a second height larger than the first height, and wherein the drive nut and the central drive screw are independently rotatable such that rotation of the drive nut is configured to expand the implant in width and rotation of the central drive screw is configured to expand the implant in height; at least one bone fastener; and at least one spinal rod attachable to the at least one bone fastener.
2. The spinal fixation system of claim 1, wherein the expandable intervertebral implant is configured to be installed through a transforaminal lumbar interbody fusion procedure.
3. The spinal fixation system of claim 1, wherein the at least one bone fastener includes a threaded shaft and a tulip head.
4. The spinal fixation system of claim 1, wherein the at least one bone fastener is a polyaxial bone screw.
5. The spinal fixation system of claim 1, wherein the at least one bone fastener is a pedicle screw.
6. The spinal fixation system of claim 1, wherein the at least one bone fastener comprises a plurality of bone fasteners configured to be installed in vertebrae adjacent to the expandable intervertebral implant.

7. The spinal fixation system of claim 1, wherein the at least one bone fastener includes a set of bone fasteners and the at least one spinal rod comprises a set of spinal rods configured to stabilize the spine.

8. A spinal fixation system comprising: an expandable intervertebral implant comprising a central drive assembly aligned along a central longitudinal axis of the implant and left and right side assemblies defining opposing lateral sides of the implant, the central drive assembly including a front plate at a distal end of the implant having a central through bore, an externally threaded central drive screw having an enlarged head and a threaded shaft positioned through the through bore in the front plate, a central actuator having a tubular body with a through bore and a pair of opposed wings, the threaded shaft of the central drive screw threadedly engaged within one end of the bore of the central actuator, a rear plate having a central through bore, an externally threaded sleeve positioned through the bore in the rear plate and threadedly engaged with an opposite end of the bore of the central actuator, a drive nut at a proximal end of the implant threadedly engaged with the threaded sleeve, and the left and right side assemblies each including upper and lower endplates, a side actuator, and a front ramp, wherein the pair of opposed wings each define a female ramp configured to receive a corresponding male ramp from each side actuator, wherein the rear plate includes a pair of female horizontal ramps defined into top and bottom surfaces of the rear plate, and each side actuator includes a pair of horizontal male ramps configured to interface with the respective female horizontal ramps of the rear plate; wherein the front plate includes a pair of female horizontal ramps defined into top and bottom surfaces of the front plate, and each front ramp includes a pair of male horizontal ramps configured to interface with the respective female horizontal ramps of the front plate, wherein the left and right side assemblies have a laterally collapsed configuration having a first width and a laterally expanded configuration having a second width larger than the first width, and wherein the left and right side assemblies have a vertically collapsed configuration having a first height and a vertically expanded configuration having a second height larger than the first height, wherein the drive nut and the central drive screw are independently rotatable such that rotation of the drive nut is configured to expand the implant in width and rotation of the central drive screw is configured to expand the implant in height; and a posterior stabilization system comprising a plurality of bone fasteners including a threaded shaft and a tulip head, and at least one spinal rod affixed to the tulip heads of the bone fasteners to stabilize a portion of the spine.

9. The spinal fixation system of claim 8, wherein the plurality of bone fasteners are configured to be inserted into pedicles of adjacent vertebrae.

10. The spinal fixation system of claim 8, wherein independent rotation of the drive nut is configured to expand the intervertebral implant in width.

11. The spinal fixation system of claim 8, wherein independent rotation of the central drive screw is configured to expand the intervertebral implant in height in-situ.

12. A method of spinal fixation comprising: inserting the expandable intervertebral implant recited in claim 1 into a disc space; expanding the expandable intervertebral implant in width and/or height; affixing a plurality of bone screws to pedicles of adjacent vertebrae; and connecting at least one spinal rod between the bone screws.

13. The method of claim 12, wherein expanding the implant in width includes independently rotating the drive nut and expanding the implant in height includes independently rotating the central drive screw.

14. The method of claim 12, further comprising, before inserting the expandable intervertebral implant into the disc space, accessing a posterior aspect of a spine, and performing a facetectomy.

15. The method of claim 12, further comprising performing a unilateral facetectomy to allow for visualization and removal of a disc.

16. The method of claim 12, wherein the expandable intervertebral implant is expanded in width to increase an overall footprint size, thereby minimizing a potential for implant subsidence.

17. The method of claim 12, wherein the expandable intervertebral implant is expanded in height in-situ to adjust lordosis.

18. The method of claim 12, wherein the step of affixing a plurality of bone screws comprises affixing a first set of bone screws is affixed and a second set of bone screws to pedicles of the adjacent vertebrae, and wherein the step of connecting at least one spinal rod comprises connecting a first spinal rod therebetween the first set of bone screws and connecting a second spinal rod is connected therebetween the second set of bone screws.
